

P02. Conflict-free Cooperation Method for Connected and Automated Vehicles at Unsignalized Intersections: Graph-based Modeling and Optimality Analysis

中英双语博士精读笔记

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Theme / 主题: Graph conflict modeling + passing order

Method family / 方法族: Conflict graph + optimal passing order / 冲突图与最优通行顺序

PDF pages / 页数: 19 **Relevance / 相关度:** 92/100

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1 论文全景速览与核心贡献 / High-Level Overview & Core Contributions

1.1 研究背景与痛点 / Background & Pain Points

这篇论文位于“Graph conflict modeling + passing order”方向。对你的 JSSP-based Intersection Management 课题，它回答的是高层调度、低层轨迹平滑、安全过滤或真实验证中的一个关键子问题：如何让车辆在路口少停、少冲突、少能耗，同时保持在线可执行。

The paper belongs to the theme of Graph conflict modeling + passing order. For a JSSP-based intersection-management thesis, it illuminates one of the key layers: scheduling, smoothing, safety filtering, or real-world validation.

Pain point / 痛点: Mostly longitudinal control, ideal communication, and no lane changing in this paper's base formulation.

1.2 核心科学问题 / Core Scientific Question

如何把无信号交叉口中复杂的相交、跟驰、合流和分流关系转化为可求解的图排序问题？

How can heterogeneous vehicle conflicts at an unsignalized intersection be encoded as graph relations that yield a conflict-free passing order?

1.3 科学贡献 / Scientific Contributions

- 方法贡献 (method contribution): Models trajectory conflicts as a conflict directed graph and coexisting undirected graph; solves passing order through improved depth-first spanning tree

and minimum clique cover methods.

- 安全/避碰贡献 (safety contribution): Conflict-free graph edges represent crossing, diverging, converging, and reachability conflicts before scheduling.
- 平滑/停止避免贡献 (smoothing contribution): Optimizes evacuation time and average delay, using distributed control to reduce idling near the stop line.
- 创新力度评判 (reviewer judgement): 较强: 图建模把交通冲突类型显式化, 便于和 JSSP 的机器冲突/工序约束建立同构关系。

1.4 核心概念对照 / Core Concepts

中文概念	English Concept	Role / 学术作用
自动交叉口管理	Autonomous Intersection Management	把信号控制替换为车辆级调度与控制。
轨迹平滑	Trajectory Smoothing	减少急加减速、停车和能耗峰值。
安全约束	Safety Constraints	把碰撞风险写成距离、时间间隔或屏障函数。
Conflict graph + optimal passing order	冲突图与最优通行顺序	该论文的核心方法族。

2 教材式学习路径与关键截图 / Textbook-Style Study Path & PDF Snapshots

2.1 学习路径 / Study Path

- 第一遍先读首页截图、摘要与引言, 确认研究问题和场景假设。
- 第二遍沿着技术路线图复现模型变量、约束和求解器输入输出。
- 第三遍逐节阅读 section deep reading, 对照 PDF 截图检查公式、图表和实验。
- 第四遍只站在审稿人视角重读实验, 判断 baseline、公平性、消融和鲁棒性。
- 最后把博士 checklist 写成自己的复现实验或论文拓展计划。

2.2 博士阅读 Checklist / Doctoral Reading Checklist

- 能否独立重写核心公式: 冲突图与排序目标 / Conflict graph and sequencing objective, 并解释每个符号的物理意义?
- 能否画出输入-模型-算法-控制-验证的数据流图?
- 能否指出至少两个强假设, 并说明它们在真实交通中何时失效?
- 能否复述实验基准是否公平, 指标是否覆盖效率、安全、能耗和计算时间?
- 能否提出一个与自己 JSSP/RL/smoother 课题直接相连的拓展实验?

2.3 关键截图讲解 / PDF Snapshot Explanation

Screenshots are selected anchors from the local PDFs for study and parallel reading. They do not replace the original paper.

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Conflict-free Cooperation Method for Connected and Automated Vehicles at Unsignalized Intersections: Graph-based Modeling and Optimality Analysis

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Abstract—Connected and automated vehicles have shown great potential in improving traffic mobility and reducing emissions, especially at unsignalized intersections. Previous research has shown that vehicle passing order is the key influencing factor in improving intersection traffic mobility. In this paper, we propose a graph-based cooperation method to formalize the conflict-free scheduling problem at an unsignalized intersection. Based on graphical analysis, a vehicle's trajectory conflict relationship is modeled as a conflict directed graph and a coexisting undirected graph. Then, two graph-based methods are proposed to find the vehicle passing order. The first is an improved depth-first spanning tree algorithm, which aims to find the local optimal passing order vehicle by vehicle. The other novel method is a minimum clique cover algorithm, which identifies the global optimal solution. Finally, a distributed control framework and communication topology are presented to realize the conflict-free cooperation of vehicles. Extensive numerical simulations are conducted for various numbers of vehicles and traffic volumes, and the simulation results prove the effectiveness of the proposed algorithms.

Index Terms—Connected and Automated Vehicle, Unsignalized intersection, Conflict-free cooperation, Distributed control

I. INTRODUCTION

Intersections are the most common merging points in urban traffic scenarios [1]. According to the Federal Highway Administration, more than 2.8 million intersection-related crashes occur in the US each year, accounting for 44% of all crashes [2]. The rapid development of the V2X technology provides an opportunity to improve traffic mobility and safety in intersection management [3]. Through vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication, a centralized controller is deployed at the intersection to coordinate the connected and automated vehicles (CAVs) to pass through the intersection; this guarantees conflict-free vehicle cooperation. Hence, CAVs have immense potential in improving traffic safety and mobility at intersections [4], [5].

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In a fully autonomous intersection scenario, a hierarchical framework is frequently used to realize the CAV coordination [6]. First, after collecting the information of the CAVs in real-time, the centralized controller schedules the arrival time of the CAVs to improve the traffic efficiency. Then, a distributed controller is applied to the CAV to implement the determined arrival plan. Multiple methods have been proposed to address the vehicle control problem, *e.g.*, fuzzy logic [7], [8], model predictive control [9], [10], and optimal control [11], [12]. Moreover, the virtual platoon method was used in [13], [14], which projects the CAVs from different lanes into a virtual lane. Thus, the typical vehicle platoon analysis methods [15] can be used in the control problem of the CAVs on different lanes as if they are traveling in the same lane.

Intersections are the converging point of traffic flows in different directions. The intersection geometry structures regulate the routes of incoming CAVs, which leads to complicated conflict relationships at intersections. Therefore, CAVs' arrival time needs to be staggered to avoid collisions. What's more, a high-efficiency conflict-free arrival plan is required to improve traffic mobility. Researchers have also pointed out that the passing order of the vehicles is the key factor influencing traffic mobility at intersections [16], [17]. As incoming CAVs approach the intersection from different directions, the optimal passing order changes constantly and the solution space increases exponentially, preventing the determination of the optimal passing order using the brute-force method. This problem is typically solved using a reservation-based method [18]. The most straightforward reservation-based solution is the first-in-first-out (FIFO) strategy, in which vehicles that enter the intersection first are scheduled to leave it first [6], [11], [19]. However, the FIFO passing order is not likely to be the optimal solution in most cases. Batch-based strategy is an improved version of the FIFO strategy, and it processes the vehicles in batches according to their direction to reduce the traffic delay [20]; however, its performance has not been fully optimized. Another widely used strategy is the optimization-based method. After formulating the scheduling problem as an optimization problem, various methods have been proposed to find the optimal passing order, *e.g.*, mixed-integer programming [21], Monte Carlo tree search [22], and dynamic programming [23]. However, the solution space exponentially

图 1: 首页与摘要 / Title and abstract page (p.1, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Strong support for representing intersection conflicts as typed graph relations, similar to the operation graph in this manuscript.

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increases with the number of the vehicles, making the use of optimization-based methods unfeasible.

Apart from these specific methods, graph theory is also introduced to find the optimal passing order of multiple CAVs, which is a discrete problem. For example, [24], [25] employs conflict graph analysis to optimize the traffic-signal phase plan. In terms of multiple CAVs scheduling, several graph-based methods have also been used. Apart from the depth-first spanning tree (DFST) algorithm proposed in [13], Petri net [26] and conflict duration graph [27] are also used in modeling the scheduling problem. However, the discussion on the optimality of the scheduling problem of multiple CAVs is inadequate.

To tackle the problem of optimality, mathematical modeling, *i.e.*, the computational reduction of the intersection scheduling problem, is an essential research aspect. Some researchers have focused on reducing this problem into typical algorithmic problems, including the job-shop scheduling problem [28], abstraction-based verification problem [29], and polling system problem [30]. However, most of these methods mainly focus on the feasibility of a conflict-free passing order solution. In practice, traffic efficiency is another critical topic that has not been fully considered in the computational reduction of this problem.

In this study, we mainly focused on finding the optimal CAV passing order at unsignalized intersections, *i.e.*, the CAV scheduling problem. Most of the existing studies have focused on obtaining the passing order through specific methods without considering the optimality and computational complexity [13], [16], [17], [19], [22]. Some have focused on the feasibility of a conflict-free solution but neglected traffic efficiency optimization [20], [26], [27], [28], [29], [30]. To address this problem, we introduce a graph-based conflict-free cooperation method to model the conflict relationship of the vehicles. Based on the conflict analysis, two novel methods are proposed to obtain the optimal vehicle passing order. Improved DFST (iDFST) obtains the local optimal solution while minimum clique cover (MCC) targets the global optimal one. The contributions of our study are as follows:

- 1) An improved DFST method is proposed based on [13] to improve traffic efficiency. Instead of considering all conflict types as one united conflict set, we categorize different conflict types to further reduce the overall depth of the spanning tree, *i.e.*, the evacuation time of vehicles. The computational complexity of the improved algorithm remains low, and the passing order solution generated by it is proved to be the local optimum.
- 2) Based on the concise and rigid graphical analysis, we first reduce the CAV scheduling problem to the minimum clique cover problem. Unlike most of the existing research [11], [13], [19], our reduction of the problem is proposed without the assumption of the FIFO principle. Hence, the global optimal passing order solution is obtained by solving the MCC problem. Moreover, we propose a heuristic method to solve the problem of numerous vehicles with a low computation burden.
- 3) A distributed feedback control method is designed to apply the scheduling results of the proposed algorithms.

In particular, the predecessor-leader following topology is used as the communication topology to lower the communication bandwidth requirement. Traffic simulations are conducted for various numbers of vehicle inputs and traffic volumes. The simulation results reveal that the proposed algorithms significantly improve both traffic efficiency and fuel economy with acceptable computational time.

The remainder of this paper is organized as follows. Section II introduces the scenario addressed in this study. Section III presents the conflict analysis, iDFST method, MCC method, and distributed control. The simulation results are presented in Section IV, and Section V concludes the paper.

II. PROBLEM STATEMENT

Figure 1(a) depicts one of the most congested intersections near the Tsinghua campus. In this scenario, because the number of approaching lanes is higher than the number of departure lanes, traffic congestion frequently occurs during rush hour. In this study, we simplified the traffic scenario by separating the vehicles into different lanes based on their destination. A hierarchical framework was established to realize the cooperation of the CAVs [6]. A *central cloud coordinator* was deployed at the intersection to guide the CAVs in driving through the intersection. After collecting the positions and velocities of all the CAVs, the upper coordinator schedules the passing order of the approaching CAVs, while the lower distributed vehicle controller controls the vehicle according to the determined passing order. The scheduling method is presented in Sections III-A, III-B, III-C, and the distributed control is presented in Section III-D. First, the following assumptions were devised to facilitate the design of the strategy.

Assumption 1: Similar to most of the existing studies on intersection traffic [11], [13], [22], lane change behavior was prohibited to guarantee vehicle safety and improve traffic efficiency. For safety reasons, vehicles with conflict relationships are not allowed to drive in the conflict zone of the intersection simultaneously.

Assumption 2: An ideal communication condition without communication delay or packet loss is under consideration. CAVs transmit their velocity and position to the central cloud coordinator through wireless communication, *e.g.*, V2I communication [31].

Assumption 3: The CAVs are capable of fully autonomous driving and are assumed to have perfect steering performance. Therefore, we only focused on the longitudinal control of the vehicle.

The incoming CAVs are indexed from 1 to N according to their arrival sequence at the control zone of length L_{ctrl} in Fig. 2, *i.e.*, only the CAVs in the control zone are under coordination. Note that since we focus on multi-vehicle scheduling problem rather than vehicle control problem, second-order vehicle model as defined in (1) is used instead of more complex

图 2: 建模或问题设定页 / Modeling or problem-setup page (p.2, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Strong support for representing intersection conflicts as typed graph relations, similar to the operation graph in this manuscript.

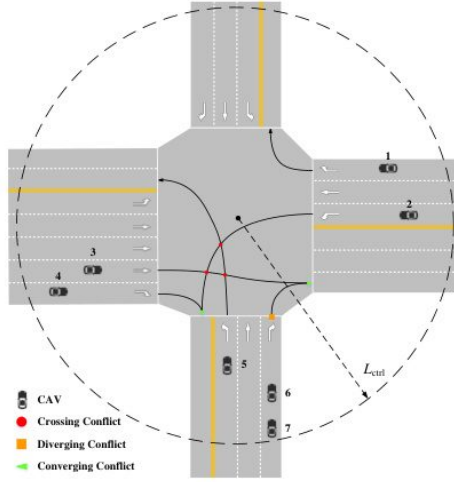


Fig. 2: Illustration of different conflict relationship between the seven CAVs. Crossing conflicts are plotted as red circles, diverging conflicts as yellow squares, and converging conflicts as green arrows.

III. METHODOLOGY

In the following paragraph, Section III-A first introduces the graph-based intersection conflict analysis method. Then, Section III-B presents the optimization of the DFST method to obtain the local optimal passing order solution, termed as the iDFST method. In Section III-C, we introduce the MCC method to find the global optimal passing order solution. Finally, the distributed controller design is presented in Section III-D to undertake the scheduling plan.

A. Conflict Analysis

Conflict analysis is one of the most fundamental area in multi-agent cooperation research [32], [33], [34]. In [13], the DFST method using a *virtual platoon* was proposed, which projects the CAVs from different lanes onto a virtual lane to avoid collision. DFST scheduling method is built on vehicle conflict relationship analysis.

In Fig. 1, the CAVs from different lanes form multiple conflict points. Despite the complicated conflict scenarios, they can be classified into the following conflict modes. Without loss of generality, several CAVs are selected in Fig. 2 to illustrate the conflict relationship. First, we introduce the route conflicts, where CAVs have trajectory intersections on their paths.

- 1) *Crossing Conflict*: Vehicles from different lanes have the potential to collide while crossing the conflict points, indicated by the 24 red circles. For example, CAV 2 and CAV 3 have a crossing conflict point.

- 2) *Diverging Conflict*: Lane changing and overtaking are not permitted, as explained in Assumption 1. Thus, vehicles on the same lane cannot pass the intersection simultaneously, as indicated by the 14 orange squares. For example, CAV 6 and CAV 7 have a diverging conflict point.
- 3) *Converging Conflict*: Vehicles from different lanes cannot enter the same lane simultaneously, as shown by the 6 green arrows. For example, CAV 2 and CAV 4 have a converging conflict point.

Apart from the above-mentioned three kinds of conflicts that describe the route conflict of the CAVs, we claim that there exists another conflict type caused by the speed and acceleration limitation of the CAVs. For instance in Fig. 2, CAV 7 arrives at the control zone when CAV 5 is about to reach the stopping line with the designed virtual platoon velocity. If we schedule CAV 5 and 7 to pass simultaneously, CAV 5 has to wait for CAV 7 near the stopping line for a long time, which jeopardizes traffic safety and efficiency. In this case, CAV 7 is not supposed to catch up with CAV 5 at the stopping line even if they have no conflict relationship, because of the limitations on the speed and acceleration of CAV 7. Hence, we further introduce the fourth type of CAV conflict, namely, the *reachability conflict*.

- 4) *Reachability Conflict*: Even if the CAVs have no route conflict, their acceleration and velocity constraints still limit them from passing the intersection simultaneously. For example, as described before, CAV 5 and CAV 7 have a reachability conflict.

Assume CAV 7 reaches the control zone at distance of L_{ctrl} with initial speed of designed virtual platoon velocity v_p . From equation (1) and (2), if CAV 7 immediately accelerates with maximum acceleration a_{max} until it reaches maximum speed v_{max} , it still takes t_{min} time until it reaches the stopping line as shown in (5).

$$t_{min} = \frac{v_{max} - v_p}{a_{max}} + \frac{1}{v_{max}} \left(L_{ctrl} - \frac{v_{max}^2 - v_p^2}{2a_{max}} \right). \quad (5)$$

We assume CAV 5 is also running with designed virtual platoon velocity v_p within the control zone, and we expect CAV 5 to maintain its speed in order to obtain a steady traffic state near the stopping line. If CAV 5 in the control zone is so close to the stopping line that it arrives at the stopping line in less than t_{min} time, CAV 7 is impossible to catch up with CAV 5. By simplifying (5), we obtain the judging condition as

$$\frac{L_{prec}}{v_p} < \frac{L_{ctrl}}{v_{max}} + \frac{(v_{max} - v_p)^2}{2a_{max}v_{max}}, \quad (6)$$

where L_{prec} is the distance of the preceding CAV from the stopping line. In other words, preceding vehicles with steady v_p form a steady scheduling plan near the intersection. If the new CAV cannot catch up with some of them because of velocity/acceleration constraints, reachability conflict is used to preserve the consistency of scheduling results. Note that most of the existing researches assume that the CAVs can reach the stopping line under all circumstances, i.e., the reachability conflict is ignored.

图 3: 核心方法或算法页 / Core method or algorithm page (p.4, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Strong support for representing intersection conflicts as typed graph relations, similar to the operation graph in this manuscript.

Note that the arrival plans are described by the virtual platoon. Therefore, typical platoon controllers [13], [15], [39] can be used in controller design. Since our research focuses on the scheduling problem rather than the control problem, a linear feedback controller is designed as follows:

$$\begin{aligned} u_i &= -\sum_{j \in \mathbb{L}_i} k_p \delta_p^{(i,j)} - \sum_{j \in \mathbb{L}_i} k_v \delta_v^{(i,j)} \\ &= -k_p \sum_{j \in \mathbb{L}_i} a_{ij} (p_j(t) - p_i(t) - D_{\text{des}}(d_j - d_i)) \\ &\quad - k_p q_{ii} (p_0(t) - p_i(t) - D_{\text{des}}(d_0 - d_i)) \\ &\quad - k_v \sum_{j \in \mathbb{L}_i} a_{ij} (v_j - v_i) - k_v q_{ii} (v_i - v_p), \end{aligned} \quad (17)$$

where k_p and k_v are the feedback gains of the distance and velocity errors of CAV i . Because we consider a homogeneous scenario, the same gains are set for all the CAVs.

As mentioned before, we consider a second-order vehicle dynamics as shown in (1). We define the car following errors as the new vehicle state.

$$\bar{\mathbf{x}}_i = \begin{bmatrix} \bar{x}_{i,1} \\ \bar{x}_{i,2} \end{bmatrix} = \begin{bmatrix} p_0 - p_i - D_{\text{des}}(d_0 - d_i) \\ v_i - v_p \end{bmatrix}, i \in \mathbb{N}^+. \quad (18)$$

The vehicle input remains the same, *i.e.*, $\bar{u}_i = u_i$.

Therefore, the car-following vehicle dynamic model is

$$\dot{\bar{\mathbf{x}}}_i = \mathbf{A}\bar{\mathbf{x}}_i + \mathbf{B}\bar{u}_i, i \in \mathbb{N}^+. \quad (19)$$

The linear feedback controller is simplified to

$$\bar{u}_i = -k_p \sum_j (l_{ij} + q_{ij}) \bar{x}_{j,1} - k_v \sum_j (l_{ij} + q_{ij}) \bar{x}_{j,2}, j \in \mathbb{L}_i. \quad (20)$$

Defining $\mathbf{k} = [k_p, k_v]^T$, we have

$$\bar{u}_i = -\sum_j (l_{ij} + q_{ij}) \mathbf{k}^T \bar{\mathbf{x}}_j, i \in \mathbb{N}^+. \quad (21)$$

IV. SIMULATION

A. Simulation Environment and Performance Index

The traffic simulation was conducted in SUMO, which is widely used in traffic researches[40]. The simulation was run on an Intel Core i7-12700F 2.1 GHz processor. The intersection scenario and lane direction settings are the same as shown in Fig. 1, and the control zone length $L_{\text{ctrl}} = 900m$. Note that CAVs should converge to their target layer when arriving at the stopping line. Therefore, adequate control zone length is provided to enable CAVs to form an equilibrium virtual platoon state. It is an interesting future research topic to analyze the shortest control zone length needed in algorithm deployment. The CAV arrival is assumed to be a Poisson distributed flow, given by

$$P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}, k = 0, 1, \dots, \quad (22)$$

where X represents the vehicle's arrival at the control zone. λ is the expected value as well as the variance of the Poisson distribution.

Traffic efficiency is the main consideration in the proposed scheduling algorithms. To better show the improvement of the

proposed algorithms, we further introduce fuel consumption as a performance indicator to evaluate the fuel economy. For the overall traffic efficiency, the evacuation time is as defined in Definition 1. In terms of individual benefits, the ATTD is considered as defined in Definition 2. As for fuel consumption, the HBFA3 model is used to evaluate the fuel consumption of the CAVs, which is integrated with SUMO [41]. Other key simulation parameters are shown in Table IV.

Remark 4: For safety concerns, D_{des} is designed relatively conservatively to avoid vehicles in different layers from collision, and vehicles are assumed to maintain desired virtual platoon speed v_p in the intersection conflict zone. Therefore, maximum speed v_{max} can only be reached in CAVs' intersection-approaching behavior. These speed settings accord with the local traffic regulations. Furthermore, it would be an interesting future direction to consider vehicle speed trajectory planning in the conflict zone and robust controller, which helps to further increase traffic efficiency and platoon performance.

B. Case Study of Algorithm Comparison

We demonstrate a case study in Example 2 to show the effectiveness of the proposed algorithms. The benchmark algorithm is the DFST method proposed in [13]. Based on DFST, the first algorithm is the iDFST presented in Section III-B. The MCC algorithm has been explained in Section III-C, which applies a heuristic algorithm to solve the MCC problem.

Example 2 (A case study of 30 CAVs): The traffic scenario remains the same as shown in Fig 1(b). A total number of 30 vehicles arrive at the intersection with $\lambda = 3$ Poisson distribution. The vehicle arrivals are generated randomly and remain identical among the three algorithms.

As mentioned before, CDG and CUG are employed to describe the conflict relationship of the vehicles. The CDG is plotted in Fig. 7(a) as explained in Definition 3. The red dashed edges are the crossing conflicts and converging conflicts, whereas the black solid edges are the diverging and reachability conflicts. Fig. 7(b) depicts the CUG defined in Definition 4, which is the complement graph of the CDG. The edges are the coexisting relationships, *i.e.*, the conflict-free vehicles.

Although identical vehicle arrivals are set for the three algorithms, different spanning trees, *i.e.*, passing order solutions, are obtained. The spanning trees are carried out by scheduling the vehicles in the same layer to pass the intersection simultaneously. Thus, the overall depth d_{all} of the spanning tree represents the evacuation time of the solution. First, the spanning trees of the DFST and iDFST methods are shown in Fig 7(c) and Fig. 7(d), which are derived from Fig. 7(a). Both the DFST and iDFST methods are based on the FIFO principle, *i.e.*, obtaining the spanning tree vehicle by vehicle. Because the DFST has limited optimization of vehicle scheduling, it is a feasible solution with $d_{\text{all}} = 14$. In contrast, iDFST separates the conflict types to find the optimal place for each vehicle and obtains a locally optimal solution. Hence, the overall depth of the iDFST method is $d_{\text{all}} = 11$. Note that the red nodes are the vehicles that are brought forward when compared with the DFST method. Fig 7(e) depicts the solution

图 4: 实验或结果页 / Experiments or results page (p.11, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Strong support for representing intersection conflicts as typed graph relations, similar to the operation graph in this manuscript.

TABLE VI: Standard deviation of computational time (seconds) comparison.

Algorithm \ Vehicle Number	10	20	30	40	50
DFST	0.007	0.012	0.021	0.031	0.048
iDFST	0.008	0.012	0.019	0.031	0.048
MCC	0.008	0.012	0.018	0.027	0.045

the traffic volumes on each lane. As before, 10 repetitions of the simulation were conducted for each number of vehicles, and the vehicle arrivals were identical across the three algorithms. The simulation results are plotted in Fig 10.

λ is the expected value of the Poisson distribution, which represents the average gap in the vehicle arrival time on each lane. As shown in Fig. 10(a), the evacuation times obtained with the three algorithms show limited fluctuations with increase in λ . The proposed two algorithms, *i.e.*, iDFST and MCC, reduce the evacuation time obtained with DFST by approximately 30%. Note that MCC has better performance when $\lambda < 3$. Recall that in Table IV, we designed the car following distance of the virtual platoon as $D_{des} = 30\text{ m}$ and the speed of the virtual leading vehicle as $v_p = 10\text{ m/s}$. Thus, the theoretical vehicle output of the intersection is bounded as $D_{des}/v_p = 3\text{ s}$ per layer. In other words, if the average input vehicle gap $\lambda < 3$, the accumulation and queuing of the vehicles is inevitable. Under this circumstance, MCC shows better performance than iDFST because of its better usage of the intersection space. Fig. 10(b) shows the ATTD performance of the algorithms. As the traffic volume increases, the vehicle is more likely to be stuck behind other preceding vehicles. Thus, the vehicle travel time is positively correlated with the traffic volume, *i.e.*, the ATTD decreases as the vehicle arrival gap increases. Similar to the evacuation time, MCC shows better performance when $\lambda < 3$. Because of the aforementioned reason for the MCC to show better utilization of the intersection space, idling behavior and fuel consumption performance are greatly improved. As shown in Fig. 10(c), iDFST saves at most 15% and MCC saves 26% of the fuel consumption when compared with DFST.

V. CONCLUSION

In this paper, a graph-based cooperation method was proposed to formulate the conflict-free scheduling problem at unsignalized intersections. Based on the graphical description of the conflict relationships among the CAVs, an iDFST method was introduced to obtain the local optimal solution. After concise and rigid graphical analysis, we reduced the CAV scheduling problem to the MCC problem, which yielded the global optimal solution. MCC problem was proven to be an NP-hard problem, therefore brute-force method to solve MCC is unrealistic in real-world deployments. To solve the problem, a heuristic method was proposed to solve MCC problem with low computation complexity in the case of a large number of vehicles. Furthermore, a distributed control framework and communication topology were designed to realize the conflict-free cooperation of the vehicles. Traffic simulations proved the

effectiveness of the proposed algorithms, where iDFST and MCC algorithms outperform the original DFST method. In the trade-off between scheduling optimality and computation efficiency, although iDFST and MCC cannot guarantee the global optimal solution, they generate rather good CAV arrival plans and have a low computational complexity of $O(N)$. Thus, both of them are qualified to schedule dozens of CAVs in real intersection scenario deployment. Note that these proposed scheduling algorithms can be applied to intersections with any geometrical structures, which proves the generalization ability of our algorithm.

A future direction of our research is to permit the lane-changing behavior, *i.e.*, the CAVs are allowed to change lanes while approaching the intersection. Several practical methods [42], [43] have been proposed to apply the lane changing behavior of multiple CAVs. It is interesting to combine these works to further extend the single intersection into multiple intersection cooperation, which will be considered in our future research. Another interesting topic is the communication topology of the virtual platoon. In this study, we designed a PLF communication topology to realize the scheduling plan. The problem of identifying the best topology is yet to be addressed. Finally, field experiments are needed for algorithm validation, where more realistic vehicle dynamic model and vehicle trajectory planning in the conflict zone should be further considered.

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图 5: 结论或局限页 / Conclusion or limitations page (p.17, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Strong support for representing intersection conflicts as typed graph relations, similar to the operation graph in this manuscript.

3 章节精读与逐段导读 / Section-by-Section Deep Reading & Walkthrough

p.1 I. INTRODUCTION

章节目标 / Learning goal: 建立研究动机：为什么传统信号控制、FCFS 或单车控制不足以处理该论文关注的交叉口协同问题。

Understand why the paper needs a new coordination method rather than a standard signal, FCFS, or single-vehicle controller.

关键论点 / Key argument: 这一部分通常把痛点落到 Graph conflict modeling + passing order：既要提高效率，又要保持安全与在线可执行。

技术焦点 / Technical focus: 精读时标出作者批判的 baseline、场景假设、交通参与者类型和隐含优化目标。

审稿追问 / Reviewer question: 作者指出的痛点是否真的需要新方法，还是可以由更强的优化器/更保守规则解决？

p.2 II. PROBLEM STATEMENT

章节目标 / Learning goal: 把自然语言交通问题压缩为变量、状态、约束、目标函数或图结构。

Translate the traffic problem into variables, states, constraints, objectives, or graph relations.

关键论点 / Key argument: 本节是复现论文的核心入口，应与公式“冲突图与排序目标 / Conflict graph and sequencing objective”一起读。

技术焦点 / Technical focus: 逐项检查车辆状态、冲突关系、控制输入、时间/空间索引和安全阈值是否定义完整。

审稿追问 / Reviewer question: 这些建模假设在混合交通、通信延迟、感知误差和高密度车流下是否仍成立？

p.4 III. METHODOLOGY

章节目标 / Learning goal: 解释算法如何把模型转化为可执行决策，并说明在线运行机制。

Trace how the paper turns the model into executable online decisions.

关键论点 / Key argument: 先依据几何轨迹和时间可达性构建冲突有向图与共存无向图；再用改进深度优先生成树和最小团覆盖等图算法求通行组与顺序；最后由分布式控制保证每辆车按序到达冲突区。

技术焦点 / Technical focus: 画出输入、核心求解器、输出和安全检查接口，明确哪些步骤是优化、哪些是规则、哪些是学习。

审稿追问 / Reviewer question: 若算法某一步失败，论文有没有 fallback、重规划或安全过滤机制？

p.11 IV. SIMULATION

章节目标 / Learning goal: 验证方法是否在公平基准下改善效率、安全、能耗或计算时间。

Check whether the claimed improvements are supported by fair baselines and meaningful metrics.

关键论点 / Key argument: 实验应与论文声称的贡献闭环: Traffic simulations over varying vehicle counts and volumes; reports efficiency and fuel-economy improvements.

技术焦点 / Technical focus: 重点追踪指标: 平均延误 (average delay)、总清空时间 (evacuation time)、吞吐量 (throughput)、燃油/能耗代理和停车次数。同时检查仿真平台、交通需求和随机性设置。

审稿追问 / Reviewer question: 实验是否只展示平均收益, 还是覆盖了高密度、异常扰动和失败案例?

p.17 V. CONCLUSION

章节目标 / Learning goal: 提炼作者承认的边界, 并把它转化为博士课题的可拓展问题。

Convert acknowledged boundaries into research opportunities.

关键论点 / Key argument: 结论不只是总结结果, 更是观察作者没有解决什么的窗口。

技术焦点 / Technical focus: 把 future work 与前文假设逐一对应, 寻找可发表的补强点。

审稿追问 / Reviewer question: 如果我是审稿人, 我会要求补哪一个实验、定理或消融?

3.1 逐段式导读 / Paragraph-Level Walkthrough

Opening motivation / 开篇动机段 先把作者的痛点翻译成一句博士问题: 如何把无信号交叉口中复杂的相交、跟驰、合流和分流关系转化为可求解的图排序问题? 读这一段时不要急着接受动机, 要追问传统 FCFS、信号灯或保守 MPC 为什么不足。

Turn the opening motivation into one doctoral-level question and test whether the paper truly needs a new method.

Assumption and setting / 假设与场景段 把车辆类型、通信条件、路口类型和动力学假设列成表。该文的关键边界包括: 车辆轨迹模板和冲突点可提前计算。; 车辆遵守分配的通行顺序, 通信近似理想。; 主要聚焦纵向控制, 横向变道不是核心问题。

List vehicle type, communication, intersection setting, and dynamics assumptions before reading the equations.

Model and notation / 模型与符号段 围绕核心公式“冲突图与排序目标 / Conflict graph and sequencing objective”复写符号表, 确认每个变量是决策变量、状态变量、参数还是约束阈值。

Rewrite the symbol table around the core formulation and classify variables as states, decisions, parameters, or constraints.

Algorithm mechanism / 算法机制段 把算法读成一条数据流: 输入交通状态, 执行 Models trajectory conflicts as a conflict directed graph and coexisting undirected graph; solves passing order through improved depth-first spanning tree and minimum clique cover methods., 输出可执行通行/控制决策, 并检查安全接口。

Read the algorithm as a dataflow from traffic state to executable sequence/control decision, with explicit safety interfaces.

Experiment and evidence / 实验证据段 不要只看作者报告的提升，要检查基准、公平性和指标。本文验证描述为：Traffic simulations over varying vehicle counts and volumes; reports efficiency and fuel-economy improvements.

Go beyond reported gains and inspect baselines, fairness, metrics, and computation time.

Reviewer synthesis / 审稿式总结段 最后把贡献拆成可发表与需补强两部分。对你课题最直接的用途是：Strong support for representing intersection conflicts as typed graph relations, similar to the operation graph in this manuscript.

Separate publishable contribution from missing evidence, then connect the result to your own thesis architecture.

4 技术路线图与贡献量评估 / Technical Route & Contribution Matrix

Step	Stage / 阶段	Content / 内容
1	输入 / Input	车辆状态、路径/车道、交通需求、信号或协同信息。
2	建模 / Modeling	冲突图与排序目标 / Conflict graph and sequencing objective
3	核心求解 / Core solver	Models trajectory conflicts as a conflict directed graph and coexisting undirected graph; solves passing order through improved depth-first spanning tree and minimum clique cover methods.
4	安全接口 / Safety interface	Conflict-free graph edges represent crossing, diverging, converging, and reachability conflicts before scheduling.
5	平滑与效率 / Smoothing and efficiency	Optimizes evacuation time and average delay, using distributed control to reduce idling near the stop line.
6	验证 / Validation	Traffic simulations over varying vehicle counts and volumes; reports efficiency and fuel-economy improvements.

Dimension / 维度	Score	Comment / 评语
理论贡献 / Theoretical contribution	4/5	冲突图与排序目标 / Conflict graph and sequencing objective
算法贡献 / Algorithmic contribution	5/5	Models trajectory conflicts as a conflict directed graph and coexisting undirected graph; solves passing order through improved depth-first spanning tree and minimum clique cover methods.

实验贡献 / Experimental contribution	3/5	Traffic simulations over varying vehicle counts and volumes; reports efficiency and fuel-economy improvements.
工程贡献 / Engineering contribution	3/5	Conflict-free graph edges represent crossing, diverging, converging, and reachability conflicts before scheduling.
博士课题价值 / PhD relevance	5/5	Strong support for representing intersection conflicts as typed graph relations, similar to the operation graph in this manuscript.

5 理论方法论深度解构 / Methodology & Theoretical Framework

5.1 问题数学建模 / Mathematical Formulation

冲突图与排序目标 / Conflict graph and sequencing objective

$$G = (V, E_d, E_u), \quad \min_{\pi} C_{\max}(\pi) = \max_{i \in V} t_i^{\text{out}} \quad \text{s.t. } (i, j) \in E_d \Rightarrow i \prec_{\pi} j$$

Symbol / 符号	含义	Meaning	Role / 建模作用
V	车辆节点集合	vehicle-node set	每辆车对应一个调度实体
E_d	有向冲突/可达边	directed conflict/reachability edges	表达必须先后通过的约束
E_u	无向共存/冲突边	undirected coexistence/conflict edges	表达是否可同批通过或需分组
pi	通行序列	passing permutation	调度算法的核心决策变量
C_max	最大清空时间	makespan / evacuation time	衡量路口整体效率

5.2 核心算法架构 / Core Algorithm Layout

先依据几何轨迹和时间可达性构建冲突有向图与共存无向图；再用改进深度优先生成树和最小团覆盖等图算法求通行组与顺序；最后由分布式控制保证每辆车按序到达冲突区。

The method builds directed and undirected graph relations from trajectory conflicts, solves groups and orders with graph algorithms, and executes the resulting order via distributed vehicle control.

5.3 收敛性、稳定性或实时性 / Convergence, Stability, Real-Time Feasibility

图约束先把不安全组合排除，控制层只需执行已筛选的顺序。实时性取决于图规模和团覆盖求解，但比直接连续优化全车轨迹更结构化。

Safety is front-loaded into graph constraints; tractability improves because the continuous-control layer receives a filtered sequence rather than solving all interactions jointly.

5.4 潜在假设与边界条件 / Assumptions & Boundary Conditions

- 车辆轨迹模板和冲突点可提前计算。
- 车辆遵守分配的通行顺序，通信近似理想。
- 主要聚焦纵向控制，横向变道不是核心问题。

6 实验验证与结果评判 / Experimental Validation & Result Evaluation

Baselines & Environment / 基准与环境：常见基准包括 FCFS、交通灯控制、无协同控制和其他图排序策略。

Validation / 验证：Traffic simulations over varying vehicle counts and volumes; reports efficiency and fuel-economy improvements.

Metrics / 指标：平均延误 (average delay)、总清空时间 (evacuation time)、吞吐量 (throughput)、燃油/能耗代理和停车次数。

Ablation / 消融：图关系本身有可解释性，但仍需更系统地拆分：仅有向图、仅无向图、不同团覆盖策略和不同交通流密度下的鲁棒性。

Reviewer reading / 审稿式阅读提醒：阅读实验时不要只看平均值；应检查交通密度、车流方向、随机种子、失败案例和计算耗时。For doctoral reading, inspect density regimes, demand patterns, random seeds, failure cases, and computation time rather than only mean improvements.

6.1 图表阅读线索 / Figure and Table Reading Cues

PDF extraction detected approximately 61 figure references and 7 table references. Exact captions remain in the original PDF; the bilingual cues below tell you what to inspect first.

图表名称	Figure/Table Name	Reading focus / 阅读重点
系统框架图	system framework diagram	先确认输入、决策层、控制层和安全约束之间的数据流。
冲突关系图 / 通行顺序图	conflict-relation or passing-order graph	把节点、边类型和先后约束翻译成 JSSP 的作业、机器和析取约束。
实验指标对比图表	experimental metric comparison plots/tables	重点看平均值之外的密度变化、失败场景、计算时间和方差。

7 审稿人视角：批判性思考与未来方向 / Critical Thinking & Future Directions

7.1 论文局限性 / Limitations & Weaknesses

- 图构造依赖精确几何和预测，异常驾驶会导致边集失真。
- 团覆盖/排序在大规模密集车流下可能成为瓶颈。
- 模型偏重全 CAV 场景，对低渗透率混合交通支持不足。

7.2 博士课题拓展点 / Research Extensions for PhD

- 把图边类型作为异构图 (heterogeneous graph) 特征，训练 GNN 调度器。
- 将最小团覆盖与 JSSP disjunctive graph 融合，建立可证明安全的调度层。
- 引入不确定边权，做分布鲁棒通行排序 (distributionally robust sequencing)。

7.3 如何服务当前课题 / Use in This Project

Strong support for representing intersection conflicts as typed graph relations, similar to the operation graph in this manuscript.

8 交互式可视化与 PDF 排版蓝图 / Interactive Visualization & PDF Layout Blueprint

动态参数 / Parameter: 冲突边密度 / Conflict-edge density, range [0.1, 0.9] .

边密度升高会减少可并行通行组，通常增大清空时间。

Higher edge density reduces parallel passing groups and increases evacuation time.

8.1 HTML 结构化布局建议 / HTML Structuring

- 顶部固定 metadata bar: 题名、作者、年份、主题、PDF 页数。
- 左侧目录/section navigator，右侧为五大精读模块。
- 公式卡片使用 `<pre>` 或 LaTeX block，紧跟符号表。
- 审稿人批注用 reviewer-note block，博士拓展用 research-idea list。
- 交互参数用 `input[type=range]` + canvas 曲线，打印 PDF 时隐藏交互控件但保留解释。

9 来源锚点与逐节阅读路径 / Source Anchors & Section-by-Section Reading Map

p.1	I. INTRODUCTION
p.2	II. PROBLEM STATEMENT
p.11	IV. SIMULATION

p.1 I. INTRODUCTION 定位问题背景、研究缺口和为什么现有保守/非协同方法不足。/ Frames the motivation, research gap, and limits of conservative or non-cooperative methods.

p.2 II. PROBLEM STATEMENT 把交通交互转写为状态、约束、目标或图结构，是精读时最应复现的部分。/ Defines states, constraints, objectives, or graph structures; this is the section to reproduce carefully.

p.4 III. METHODOLOGY 给出算法流程和求解机制，应重点追踪输入、输出和安全接口。/ Gives the algorithmic mechanism; track inputs, outputs, and safety interfaces.

p.11 IV. SIMULATION 检验方法是否真的改善效率、安全或能耗；要关注基准是否公平。/ Tests efficiency, safety, or energy claims; inspect whether baselines are fair.

p.17 V. CONCLUSION 总结贡献和未来方向，也常暴露作者承认的边界条件。/ Summarizes contributions and often reveals author-recognized boundaries.